

Section - B

Q.2(i)

A:2(i)

~~Nuclear~~ chemistry

Industrial chemistry

i) Nuclear chemistry is the branch of chemistry that studies the change of nuclei in an atom, their radioactive decay etc.

Example :-

Uranium-238

Industrial chemistry is the branch of chemistry concerned with the large scale production of chemical for mankind.

Example :-

ammonia by the haber process.

Q.2(ii)

A:2(ii)

Property	Solid	Liquid	Gas
Density	high	high	low
Compressibility	not compressible	moderate	high
Ability to flow	Cannot flow	Can flow	Yes.

Q: 2 (iii) (or)
A: 2 (iii) (or)

Isotopes:-
Isotopes are the atoms of an element has same atomic number but different mass numbers.

Isotope of hydrogen :-

Isotope	Protium	Deuterium	Tritium
Proton	1	1	1
Symbol	H_1^1	H_1^2	H_1^3
Electrons	1	1	1
Mass number	1	2	3
Atomic number	1	1	1
Neutron	0	1	2

Tritium (H_1^3) is a radioactive.

Q: 2 (iv)

A: 2 (iv)

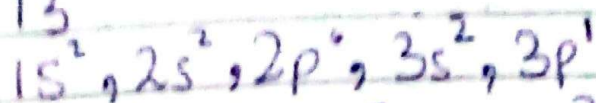
Na : 11

$1s^2, 2s^2, 2p^6, 3s^1$

Group : 1

Period : 3rd

Ex. AL: 13



Group: $3 + 10 = 13$ or 3

Period: 3rd.

Q: 2(v)(or)

A: 2(v)(or)

Ionic bond is the strong electrostatic force of attraction b/w oppositely charged ions (positively charged metal cation and negatively charged non-metal anion). It forms when an electropositive metal transfers electrons to an electronegative non-metal.

Deriving the formula of CaCl_2 :-

- 1) Identify the ions:- Ca^{2+} (charge = +2)
 Cl^{-} (charge = -1)
- 2) Compound must be electrically neutral
To balance +2, we need two Cl^{-} ions.
- 3) Formula:- CaCl_2 derived.

Q: 2(vi)(or)

A: 2(vi)(or)

molar mass of $\text{C}_{12}\text{H}_{22}\text{O}_{11}$:-

$$12(12) + 22(1) + 11(16)$$

$$= 144 + 22 + 176$$

$$= 342 \text{ mass (g)}$$

mass = molar mass $\times$ M.M.

$$2.5 \times 342 = 855 \text{ g}$$

Average N.O of molecules:-

$$\frac{2.5 \times 6.022 \times 10^{23}}{1.506 \times 10^{24}}$$

S: 2(vii) (or)

A: 2(vii) (or)

moles of H_2O

① molar mass of H_2O :- $1(2) + 1(16)$
 $2 + 16$

$$18 \text{ g/mol}$$

mass :- 100 g

moles :- $\frac{\text{G.M.}}{\text{M.O.M.}}$

$$\Rightarrow \frac{100}{18}$$

$$= 5.56 \text{ mol/g}$$

② molar mass of H_2 :- $1(2)$
 2 mol

mass :- 5

moles :- $\frac{\text{G.M.}}{\text{M.M.}}$

$$= \frac{5 \text{ g}}{2 \text{ mol}} = 2.5 \text{ mol/g}$$

Q: 2 (viii) (or)

A: 2 (viii) (or)

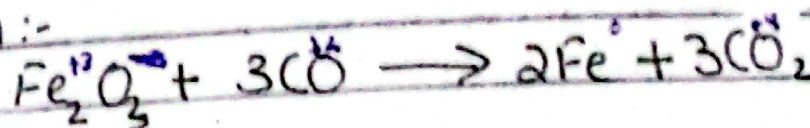
Oxidizing agent:-

A substance that causes another substance to be oxidized by accepting electrons from it. The oxidizing agent is reduced itself.

Reducing agent:-

A substance that causes another substance to be reduced by donating electrons to it. The reducing agent is oxidized itself.

Equation:-



Fe :- +3 $\rightarrow$ 0 (reduced) (oxidizing agent)

C :- +2 $\rightarrow$ +4 (oxidized) (reducing agent)

Q: 2 (ix) (or)

A: 2 (ix) (or)

The solution changes from purple to colourless. Potassium permanganate (KMnO_4) is a strong oxidizing agent. When added to a reducing agent (such as FeSO_4), the MnO_4^- ions (purple) are reduced to Mn^{2+} ions (colourless), which discharges purple ~~color~~ color.

The oxidation number of Mn changes from +7 (in MnO_4^-) to +2 (in Mn^{2+}) a decrease of 5. This represents 5 electrons.

Q: 2(n)

A: 2(n)

Enthalpy of reaction:-

is the amount of heat or thermal energy evolved or absorbed when a chemical reaction takes place :- It is denoted by Δ

Exothermic reaction:-

ΔH is negative (heat released to surrounding).

Endothermic reaction:-

ΔH is positive (heat absorbed around ~~releasing~~ surrounding).

Q: 2(n) (or)

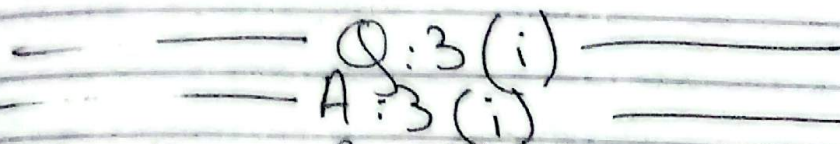
A: 2(n) (or)

3 conditions for equilibrium:-

- ① The concentration of reactants or products remain unchanged.
- ② The temperature of the system remains constant.
- ③ The pressure or volume of the system remains constant.

Dynamic equilibrium:-

means that both the forward and reverse reactions are still occurring at the same rate, the system is not static. The concentration of reactants and products remain constant because the rates of the 2 opposing reactions are equal.



Atomic theory for John Dalton:-

- ① All matter is composed of tiny indivisible particles called atoms.
- ② Atoms particular to an element are identical in mass properties.
- ③ Atoms can neither be ~~created~~ created nor be destroyed.

① $Z = 19$, $K = 39$
protons = 19
electrons = 19
neutrons = 20

② E.C :- $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1$
group :- 1
period :- 4th

$$\frac{Q.3(ii)(or)}{A:3(ii)(or)}$$

① Empirical formula:

is the simplest whole number ratio of atoms in a compound.

Molecular formula:

is the formula that gives the actual number of each type of atoms in one molecule of the compound.

① Numerical:-

① molecular formula:- CH

$$12 + 1$$

13 amu.

$$n = \frac{\text{molecular mass}}{\text{molecular formula}}$$

$$= \frac{78}{13} \Rightarrow 6$$

② Atoms in 1.25 mol Zn and in 0.2 mol AL.

$$\begin{aligned} \text{① Atoms in 1.25 mol Zn :-} \\ &= 1.25 \times 6.022 \times 10^{23} \\ &= 7.528 \times 10^{23} \text{ atoms} \end{aligned}$$

Atoms ~~of~~ in 0.2 mol Al :-

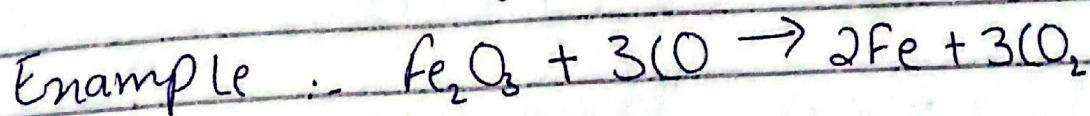
$$0.2 \times 6.022 \times 10^{23}$$

$$1.2044 \times 10^{23}$$

Q: 3(iii)
A: 3(iii)

① In terms of oxygen:-

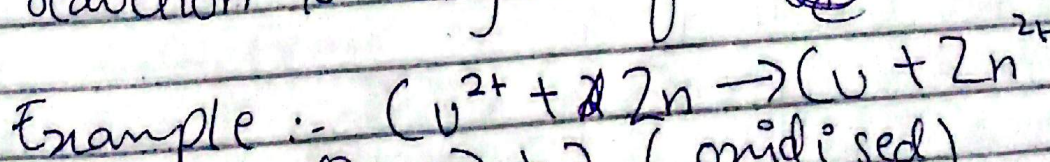
Oxidisation is the gain of oxygen and
reduction is the ~~gain~~ ^{loss} of oxygen.



CO gain oxygen (~~red~~ oxidised)
 Fe_2O_3 lose oxygen (reduced)

② Terms of electrons:-

Oxidation is the loss of ^{electrons} ~~oxygen~~ and
reduction is the gain of ~~oxygen~~ ^{electrons}

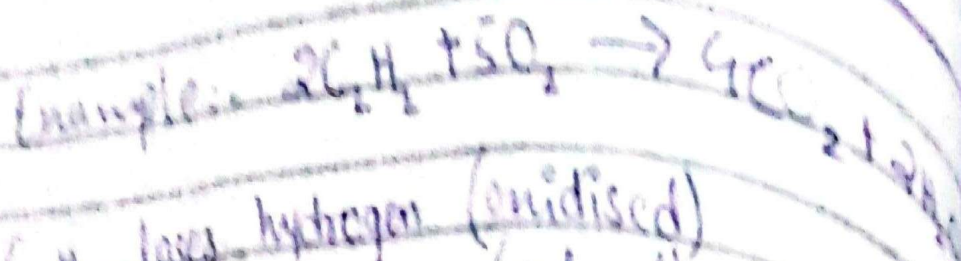


Zn :- 0 $\rightarrow$ +2 (oxidised)

Cu = +2 $\rightarrow$ 0 (reduced)

③ Terms of hydrogen :-

Oxidation is the loss of hydrogen and
reduction is the gain of hydrogen.



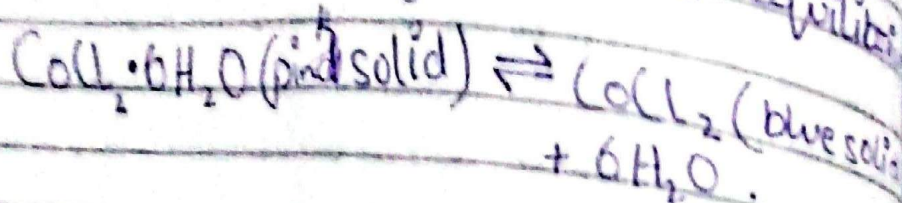
C_2H_2 loses hydrogen (oxidised)
 O_2 loses gains hydrogen (reduced)

Q: 2(iv)(or)

A: 3(iv)(or)

Hydrated cobalt (II) chloride and
 anhydrous cobalt (II) chloride

exist in the following reversible equilibrium



On heating:

Temperature increases,
 equilibrium shifts to the right (forward
 reaction is ~~exothermic~~ endothermic,
 absorbs heating to reduce the change.
 Hydrated cobalt (II) chloride (pink) loses
 water and becomes anhydrous cobalt (II)
 chloride (blue).

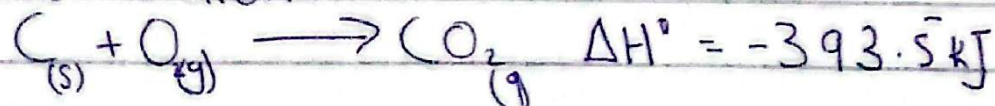
System :-

The part of the universe
 under study in a chemical reaction
 e.g. reactants and products in a test tube

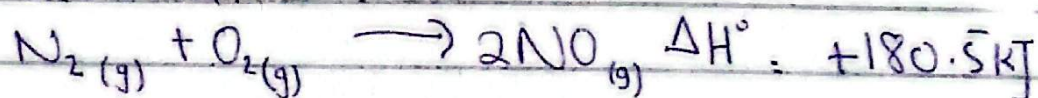
Surroundings:-

Everything else in the universe outside the system (e.g. test tube, water, bath and the air around it).

Exothermic reaction:-



Endothermic reaction:-



MCQs:-

1. B
2. C
3. B
4. C
5. C
6. C
7. C
8. B
9. B
10. B
11. B
12. D